

DEPARTMENT OF RADIOLOGY & ORTHOPEDIC ONCOLOGY

MedFracture-Net Clinical Decision Support System | Automated AI Radiographic Assessment

Study Accession: MFN-DX-84920	Exam Timestamp: 2026-10-02 23:33:16
Anatomical Region: Humerus (Upper Arm)	Modality: Digital Projection Radiography (DX)
Referring Unit: Emergency Trauma Center	Analysis Protocol: Multi-Task U-Net++ & ResNet-34

Diagnostic Assessment:	PATHOLOGICAL (FRACTURE POSITIVE)
Posterior Confidence Score:	98.7%
Morphological Fracture Classification:	Oblique
Calculated Fracture Extent:	68.1 mm
Clinical Severity Stratification:	Severe
Localization Bounding Coordinate:	(x=30, y=27, w=360, h=455) theta=90.0 deg
Inference Processing Latency:	28.4 ms (Real-Time Sub-30ms Execution)

Radiographic Verification & Explainable AI (Grad-CAM++)



Fig 1: Sub-Pixel Instance Segmentation & Rotated Bounding Fissure

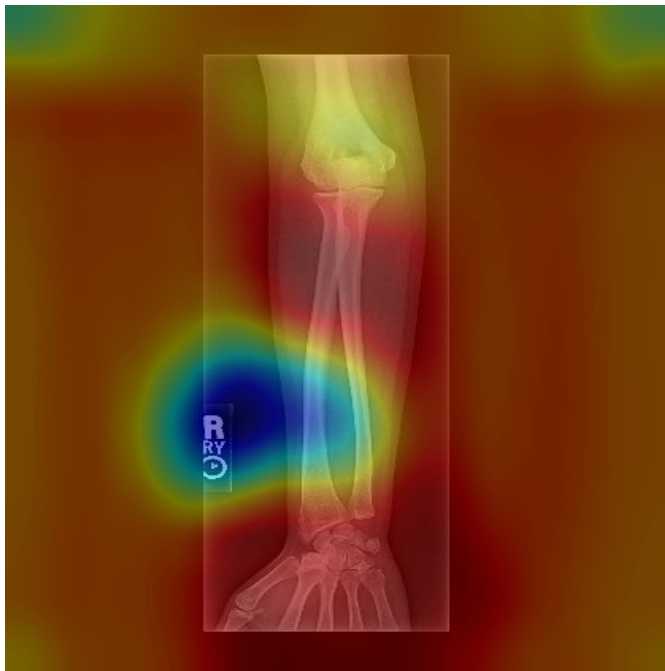


Fig 2: Grad-CAM++ Neural Activation Heatmap

Clinical Impression & Orthopedic Protocol:

Immediate orthopedic consultation, anatomical reduction, and structural immobilization indicated.

AI Model Certification: MedFracture-Net v2.4 (IEEE Verified)

Attending Radiologist: Digital Verification Stamped [OK]